8.4 Proportionality Theorems

Name _____ Period ____ Date _____

Geometry 8.4 (30-minute plan) · Student sheet

I can…

- use a segment parallel to one side of a triangle to find a missing length.
- decide whether a segment is parallel to a side of a triangle.
- use parallel lines and angle bisectors to find missing lengths.

Vocabulary · fill in the missing word

Triangle Proportionality Theorem — a line _____ to one side of a triangle that crosses the other two sides divides them proportionally

Converse — if a line divides two sides of a triangle _____, it is parallel to the third side

Transversal — a line that _____ two or more lines

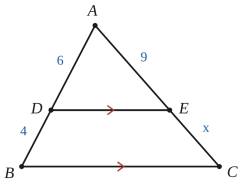
Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Notes and examples (fill in each note, then work the example)

Triangle Proportionality Theorem: if a line _____ to one side of a triangle intersects the other two sides, then it divides the two sides _____.

Example 1

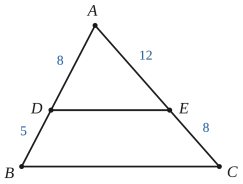
In △ABC, $\overline{DE} \parallel \overline{BC}$. AD = 6, DB = 4, and AE = 9. Find EC.



Converse: if a line divides two sides of a triangle proportionally, then it is _____ to the third side.

Example 2

In △ABC, AD = 8, DB = 5, AE = 12, and EC = 8. Is $\overline{DE} \parallel \overline{BC}$?

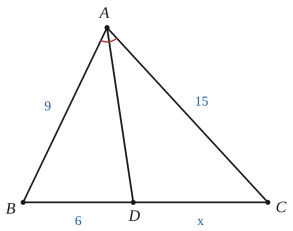


Three Parallel Lines Theorem: if three parallel lines intersect two _____, then they divide the transversals proportionally.

Triangle Angle Bisector Theorem: if a ray bisects an angle of a triangle, then it divides the opposite side into segments proportional to the other two _____.

Example 3

$\overline{AD}$ bisects $\angle BAC$. AB = 9, AC = 15, and BD = 6. Find DC.



Part 2 · Partner practice

Partner practice

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. AD = 8, DB = 12, and EC = 15. Find AE.

